Why LLM Math RL Is Not AlphaZero

AlphaZero keeps sampling a renewable 50/50 frontier. Fixed LLM math datasets collapse into all-right or all-wrong prompts, so the useful information integral is finite.

rollout/path information integral | finite LLM projection | watermelon -> seeds | transfer geometry of prompts

Core logic

AlphaZero-like self-play

```
p_t(win) ~= 1/2
I_T = sum_t H_b(p_t) ~= T
No fixed dataset cap as T grows
```

Fixed LLM math RLVR

```
P_i = sum_t H_b(p_i(t))
G_i = sum_t H_b(p_i(t)) Delta p_i^+
0 <= G_i <= 1 bit per prompt
```

Dataset geometry

```
T_i = [DeltaP_i1, ..., DeltaP_iN]
d(i,j) = 1 - cos(T_i, T_j)
Joint useful mass <= sum of parts
```

1. Traditional RL: the information integral can keep growing

Renewable 50/50 frontier

Connect4 evidence

The important AlphaZero fact is not merely that the reward is terminal and binary. Self-play changes the rollout distribution itself. The current agent keeps meeting opponents and positions near its own ability, so roughly half the rollouts can remain positive and half negative.

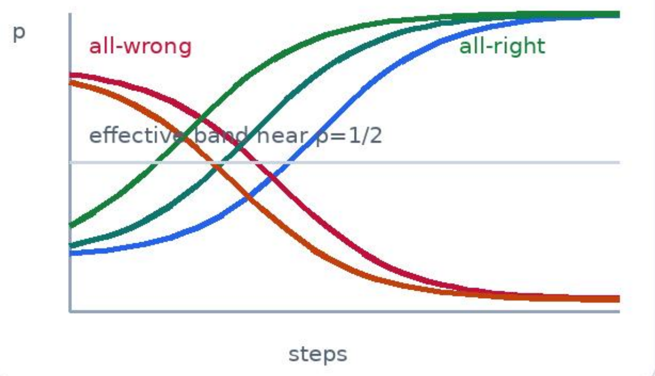
Therefore the accumulated information integral is controlled by continuing interaction and update budget, not by a fixed list of questions.

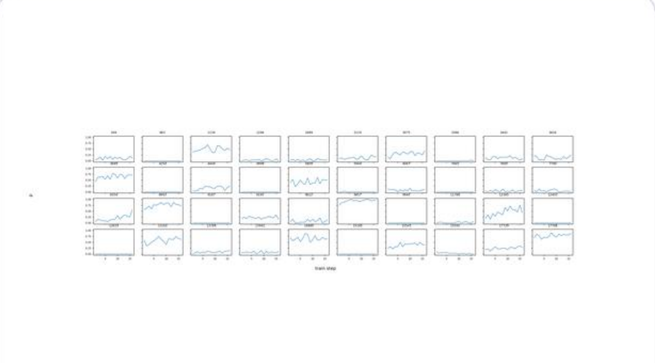
SELF-PLAY VS RANDOM	SELF-PLAY VS WEAK	SHAPED VS WEAK
0.886	**0.625**	**0.750**

Poster takeaway: self-play renews the frontier. It can keep creating informative rollouts instead of depleting a finite source.

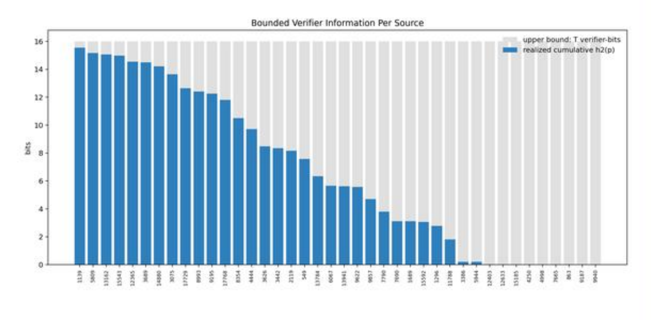
2. LLM math RLVR: fixed questions have finite useful information

Fixed math prompts collapse

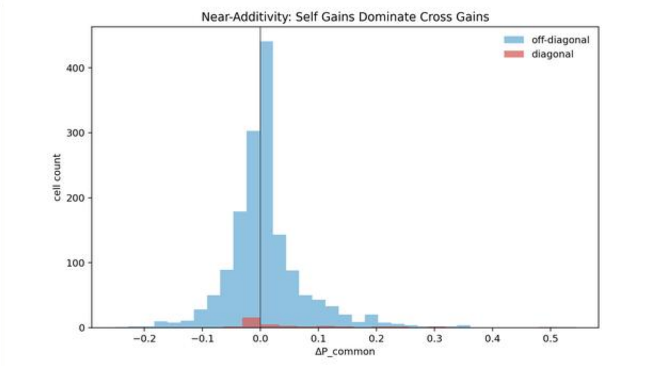




Per-prompt p_i(t) trajectories.



Cumulative Bernoulli projection.



Self transfer dominates off-diagonal.

Per-prompt finite score

```
H_b(p) = -p log2 p - (1-p) log2(1-p)
G_i = sum_t H_b(p_i(t)) Delta p_i^+(t)
0 <= G_i <= 1 bit
```

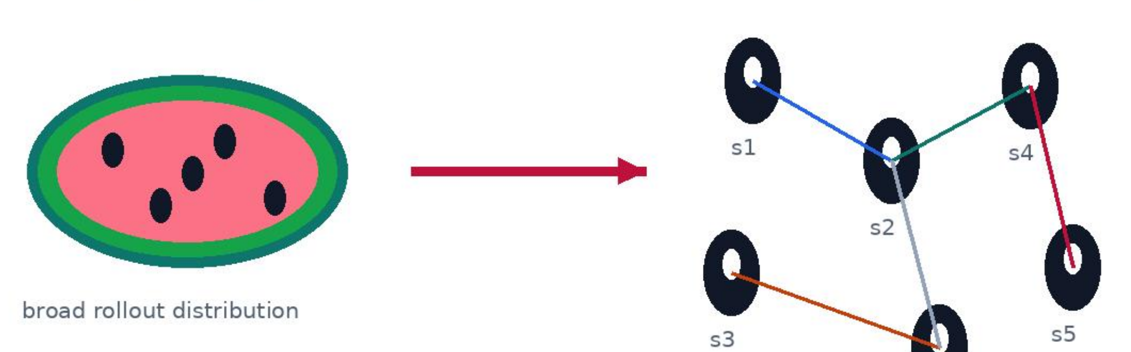
A math prompt often leaves the useful middle band: the model either solves it reliably or fails it reliably. Once rollout rewards are all 1 or all 0, the binary verifier carries little local contrast. Over a fixed dataset, each question has its own cap, and the joint cap is usually smaller than the sum because update directions and transfer fingerprints are not perfectly aligned.

USED PROJECTION	DIAG DELTAP	OFFDIAG DELTAP	OFFDIAG / DIAG	ADDITIVITY
41.5%	**0.0706**	**0.0091**	**0.129**	**0.871**

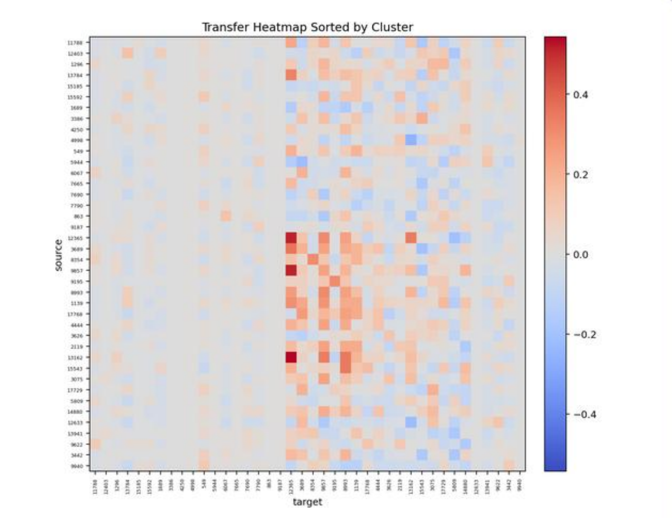
This is the poster's bounded-information evidence: not h2(p) as the whole definition, but the finite Bernoulli projection of a rollout/path integral on a fixed verifier dataset.

3. What RL actually does: shrink a distribution into problem seeds

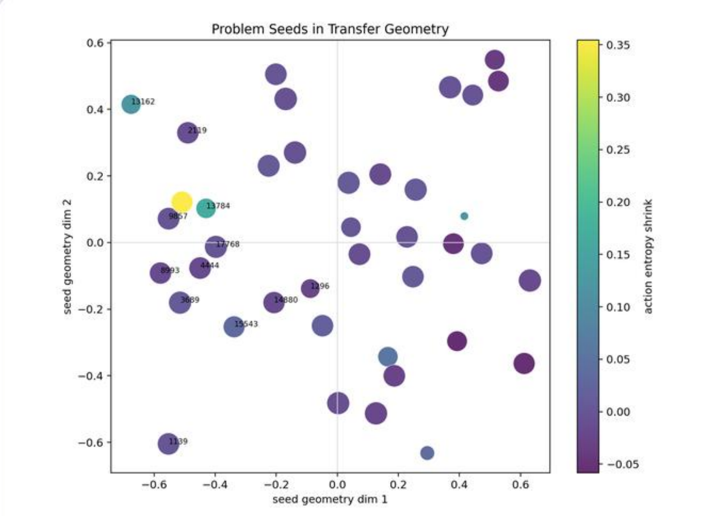
RL sharpening: watermelon -> seeds



A source prompt is represented by its transfer fingerprint T_i = [DeltaP_i1, ..., DeltaP_iN]. Distances between fingerprints turn a fixed dataset into a geometry.



Directed transfer matrix.



Problem seeds embedded by transfer fingerprints.

Dataset distance

```
DeltaP_ij = p_j(after train i) - p_j(base)
T_i = [DeltaP_i1, ..., DeltaP_iN]
d(i,j) = 1 - cos(T_i, T_j)
```

The same bounded-integral story gives a practical tool. Train on source i, evaluate target j, and use the directed transfer matrix as a map of the dataset. Strong diagonal dominance means the dataset is close to additive; sparse off-diagonal edges reveal reusable skills, clusters, and possible curriculum shortcuts.

Research program

Measure the rollout/path information integral, identify when fixed verifier datasets run out of useful middle-band signal, and use transfer geometry to allocate prompts, renew the frontier, and test whether LLM math RL can become less data-additive.